Pranjal Shrivastava

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Education

- **San Jose State University** — San Jose, CA
  M.S. Software Engineering, Data Science Specialization | GPA 3.7/4.0 — *Aug 2025 - May 2027 (Expected)*
  Relevant Coursework: Machine Learning, Deep Learning, Distributed Systems, Cloud Computing, Data Mining
- **Lakshmi Narain College of Technology and Science** — India
  B.Tech, Computer Science and Engineering | CGPA 9.2/10 — *Aug 2019 - Jul 2023*
  Relevant Coursework: Data Structures and Algorithms, Operating Systems, Computer Networks, Database Management Systems

Professional Experience

- **Work4Flow Inc.** — May 2026 - Sep 2026
  Software Engineer Intern — *San Jose, CA*
  - Built a **multi-agent orchestration service** in **Python** (planner, retriever, writer, and evaluator agents) that decomposes support data into multi-step tasks to generate knowledge articles, cutting turnaround **98%** (2 days to 20 min).
  - Designed a **tool-calling layer** with typed JSON-schema **structured outputs** across **Claude** and **GPT-4o**, adding retries, validation, and fallbacks that reduced malformed tool calls from **11%** to under **0.5%**.
  - Implemented **short- and long-term agent memory** (Redis session state, pgvector store, context summarization) and served the runtime as **FastAPI microservices**, cutting prompt tokens **43%** and raising edit-free acceptance from **58% to 91%**.
- **Accenture** — Oct 2023 - Aug 2025
  Associate Software Engineer
  - Developed **Java (Spring Boot)** and **Python** backend microservices and **REST/gRPC APIs** for an enterprise client platform serving **3M+ requests/day** at **99.95%** availability on Azure Kubernetes Service.
  - Engineered a **generative AI** service that converts plain English compliance rules into Azure Policy definitions via LLM **structured outputs**, validating every output through automated checks and cutting authoring time **70%**.
  - Built **real-time data pipelines** with **Kafka** and **PySpark** over **SQL (PostgreSQL)** and **NoSQL (Cosmos DB, MongoDB)** stores, redesigning partitioning and indexes to cut processing time **28%** and end-to-end latency **15%**.
  - Optimized a latency-critical **C++** data-validation module with memory pooling and multithreading, improving throughput **3.2x** for regulated data migrations that reached **100% recovery** with no lost records.
  - Implemented **security and privacy** controls (OAuth2/mTLS, secrets rotation, PII masking) and automated **CI/CD** in Azure DevOps, cutting pipeline run time **18%** and incident MTTR **35%**.

Projects

- **EdgeMind - Hybrid Edge-Cloud Agent Orchestration Runtime** — Python, C++, gRPC, Redis, ONNX Runtime, Docker
  - Built an **agent orchestration runtime** supporting **tool use**, **memory management**, and **multi-step reasoning**, with a planner that routes each step to an **on-device SLM** or a **cloud LLM** based on latency, cost, and privacy.
  - Kept **68%** of requests on-device, cutting cloud inference cost **54%** and median response latency from **1.9s to 620ms** while keeping task success within **2%** of a cloud-only baseline on a 500-task benchmark.
- **On-Device Inference Engine for Laptop AI Copilots** — C++, PyTorch, ONNX Runtime, TensorFlow Lite, CoreML
  - Exported **PyTorch** vision and language models to **ONNX Runtime**, **TensorFlow Lite**, and **CoreML** with **INT8 quantization**, shrinking model size **4x** with under **1.5%** accuracy loss.
  - Wrote a **C++** inference server with batching and NPU/GPU execution providers, reaching **38 tokens/s** on a consumer laptop and lowering peak memory **47%**, exposed to apps through a local **gRPC API**.
- **StreamSense - Real-Time Distributed AI Inference Pipeline** — Java, Kafka, Flink, PostgreSQL, MongoDB, Kubernetes
  - Built a **streaming inference pipeline** in **Java** that ingests device telemetry through **Kafka** and **Flink**, runs model scoring, and persists results to **PostgreSQL** and **MongoDB** at **50K events/s** with **p99 latency under 120ms**.
  - Added **privacy-preserving** on-device PII redaction and end-to-end encryption, and autoscaled **Kubernetes** workers on lag, cutting compute cost **31%** at peak load.

Technical Skills

- **Languages:** Python, Java, C++, Go, SQL, JavaScript
- **LLM & Agents:** Multi-Agent Systems, Agent Orchestration, Tool Use / Function Calling, Memory and Context Management, Task Decomposition, Structured Outputs, Prompt Engineering, RAG, MCP, LangGraph, OpenAI API, Anthropic Claude API
- **Backend & Distributed Systems:** Microservices, REST/gRPC APIs, FastAPI, Spring Boot, Kafka, Flink, Redis, Async Job Queues, Distributed AI Workloads, Cloud-Edge Hybrid Architectures
- **AI Deployment & On-Device:** PyTorch, TensorFlow Lite, ONNX Runtime, CoreML, INT8 Quantization, vLLM, Model Serving, Real-Time Inference Optimization
- **Databases:** PostgreSQL, MySQL, MongoDB, Cosmos DB, Redis, pgvector
- **Cloud, DevOps & Security:** Azure, AWS, GCP, Docker, Kubernetes, GitHub Actions, Azure DevOps, OAuth2, mTLS, PII Masking, Agile/Scrum